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# POPULATION-ADAPTIVE SPEECH ANALYSIS FOR EQUITABLE AD/ADRD DETECTION

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Team Harris and Kielo

## 1 Overview

The ideal acoustic ADRD detection tool would process any short speech clip and accurately assign a diagnosis, making early ADRD detection as straightforward as diagnosing a broken bone from an X-ray. As a small contribution towards this ambitious goal, we build on insights from our Phase 2 submission to improve its clinical potential by creating a model that is:

1. Explainable - diagnosis include factors communicable to practitioners, patients, and families
2. Generalizable - adapts to new populations while incorporating expert knowledge

To do this, we refactor our model around the three basic information types in clinical speech data:

1. What is being said? (Linguistic)
2. How is it being said? (Paralinguistic)
3. Who is saying it? (Demographics/population)

Our approach identifies novel linguistic and paralinguistic factors while adjusting for demographic and sampling effects. This approach results in interpretable building blocks for further research while maintaining predictive power equivalent to our Phase 2 approach.

We leverage two key external resources: the Mozilla Open Voice dataset to learn healthy speech patterns (enabling us to identify ADRD patients as outliers), and a curated synthetic dataset to quantify previously difficult-to-measure linguistic factors. This synthetic dataset was built using AD/ADRD research input to generate transcripts via ChatGPT without submitting patient data.

## 2 Performance

### 2.1 Clinical Effectiveness and Equity Improvements

Our refined model achieves 78.0% accuracy in identifying patients (a +3% improvement over our Phase 2 submission). While this falls short of diagnostic-level precision required for definitive diagnosis, it represents clinically meaningful performance for a screening tool that could help prioritize patients for specialist evaluation.

The most significant advance from Phase 2 is improved equity across patient populations. Table 1 demonstrates that our approach to explicitly modeling population differences has substantially reduced performance disparities, at the cost of our new model assuming we have access to reliable population base rates.

### 2.2 Key Equity Insights

**Spanish-speaking patients:** The most dramatic improvement occurred for Spanish speakers, where performance increased from 0.69 to 0.85 AUC, performing nearly as well as for English speakers, which is our majority data class.

Table 1: Performance equity improvements across patient populations (AUC scores)

| Category       | Subcategory                     | Phase 3 AUC | Phase 2 AUC | N    |
|----------------|---------------------------------|-------------|-------------|------|
| Age            | Young (69 and below)            | 0.87        | 0.87        | 522  |
|                | 70 to 79                        | 0.90        | 0.84        | 606  |
|                | 80 and Older                    | 0.87        | 0.78        | 930  |
| Education      | Incomplete Secondary            | 0.86        | 0.75        | 262  |
|                | Completed Secondary             | 0.86        | 0.76        | 488  |
|                | Higher Education                | 0.88        | 0.83        | 580  |
| Gender         | Female                          | 0.88        | 0.84        | 1219 |
|                | Male                            | 0.88        | 0.81        | 839  |
| Language       | English                         | 0.88        | 0.84        | 1655 |
|                | Spanish                         | 0.85        | 0.69        | 360  |
| Race           | White                           | 0.88        | 0.76        | 99   |
|                | African American                | 0.93        | 0.92        | 9    |
| Intersectional | Young, Female                   | 0.84        | 0.89        | 311  |
|                | Young, Incomplete Secondary Ed. | 0.88        | 0.85        | 55   |
|                | Young, Spanish Language         | 0.74        | 0.67        | 75   |

**Older adults:** Performance for patients 80 and older improved substantially (0.78 to 0.87), addressing a critical gap since this population has the highest AD/ADRD risk.

**Education levels:** Patients with less formal education showed marked improvement, reducing the risk that educational background would be confused with cognitive decline.

**Young, Spanish-speaking:** Patients continue to under-perform, highlighting a niche that may be underrepresented in the data.

### 2.3 Clinical Utility Framework

Our new model provides a confidence estimate with each prediction, which automatically stratifies patients into three actionable categories:

- **High confidence screening (27% of cases):** Performance sufficient for initial triage
- **Decision support (43% of cases):** Useful additional information for clinical assessment
- **Specialist referral (30% of cases):** Complex cases requiring detailed neuropsychological evaluation

This framework ensures that clinical resources are directed toward patients who most need expert assessment, while providing useful screening for most cases.

## 3 Methodology

### 3.1 Addressing Core Clinical Challenges

Traditional speech-based screening faces three major obstacles in clinical practice: (1) population differences - what sounds "normal" varies dramatically between a 65-year-old Spanish speaker and an 85-year-old native English speaker, (2) interpretability - clinicians need to understand why a model flagged a patient, not just that it did, and (3) dataset bias - research datasets often don't represent real clinical populations.

Corpus-level analysis of phase 2 results revealed that ADRD vs. MCI distinctions were driven by corpus rather than genuine feature differences. Since MCI represents a partial progression towards ADRD, we modify the problem to be binary classification while maintaining acceptable false negative rates.

### 3.2 1. Demographic and Population Effects

**Clinical insight:** Age, education, and background fundamentally shape speech patterns, but existing models often confuse these natural variations with disease markers.

**Our approach:** Rather than treating all patients identically or creating separate models for each group, we use "partial pooling" - learning population-specific scaling factors while sharing core disease patterns across groups. This means insights from English speakers inform Spanish speaker assessments, but with appropriate influence.

**Key finding:** When deploying to a new population, the model adapts by applying a local baseline expectations (requiring only the local AD/ADRD prevalence rate, which can be provided by epidemiological studies or clinical experts) while preserving proven disease detection patterns.

### 3.3 2. Voice Quality and Acoustic Patterns (Paralinguistic)

**Clinical insight:** AD/ADRD affects motor control, breathing, and cognitive processing in ways that alter voice quality, speaking rhythm, and prosody.

**Our approach:** We use GeMAPS, a well-established set of 88 acoustic features that capture voice quality, pitch patterns, and timing. However, unlike language analysis where powerful pre-trained models exist, no equivalent "foundation models" exist for voice quality analysis. This meant we had to establish our own healthy baseline by training on 150,000+ speech samples from Mozilla's Open Voice dataset. Learning rich representations from healthy data allows more nuanced feature extraction than traditional approaches - analogous to analyzing a complex curves and shapes of an image rather than just its average pixel color.

**Key finding:** Voice quality changes were the strongest predictor in our model. This suggests that acoustic changes may be detectable earlier than previously recognized in AD/ADRD progression.

### 3.4 3. Language Content and Structure (Linguistic)

**Clinical insight:** AD/ADRD affects language in subtle but measurable ways - reduced vocabulary diversity, simpler sentence structures, and difficulties maintaining conversational coherence. However, many clinically recognized symptoms are nearly impossible to measure computationally with traditional tools.

**The measurement challenge:** While computers can easily count words or measure sentence length, they struggle with complex symptoms like "spatial confusion" (inconsistent location references) or "conceptual fixation" (becoming stuck on single aspects of a scene). Traditional natural language processing works well for simple metrics but fails to capture the nuanced linguistic patterns that clinicians recognize.

**Our approach:** We solved this by creating targeted training data for 66 specific AD/ADRD linguistic symptoms. Using a two-step process:

1. **Scenario generation:** We created 100 new "Cookie Theft"-style scenarios - complex scenes with multiple actions and relationships that parallel the original assessment tool
2. **Symptom modeling:** For each of the 66 linguistic markers, we generated paired examples showing how someone with healthy cognition versus someone exhibiting that specific symptom would describe the same scene

This created a comprehensive training dataset where we could teach classifiers to recognize specific patterns like "pronoun overuse without clear antecedents" or "systematic avoidance of challenging semantic categories."

**Research implications:** This \$25 approach (released with MIT license on HuggingFace) provides a template for measuring any linguistic symptom that experts can describe but computational tools cannot quantify. The process could be enhanced by collaborating with neuropsychologists to validate and strengthen the exemplar data.

**Key findings:** The model identified several clinically meaningful patterns:

- **Pronoun overuse:** Patients with cognitive decline used excessive pronouns without clear antecedents - saying "it" and "that" when healthy speakers would use specific nouns
- **Conceptual fixation:** Difficulty shifting attention across scene elements, becoming stuck on single aspects rather than providing comprehensive descriptions
- **Adaptive circumlocution:** Interestingly, flexible avoidance patterns actually indicated *healthier* cognition - patients who could adaptively switch between different explanatory strategies performed better than those with rigid linguistic patterns. This likely reflects healthy individuals' ability to jump between different activities in the Cookie Theft scene, representing an experimental artifact that could be improved in future research.

### 3.5 Model Integration

These three components work together using a statistical framework that naturally quantifies uncertainty. When voice quality clearly indicates impairment, the model expresses high confidence. When only subtle linguistic changes are present, confidence drops appropriately, triggering human review.

**Clinical workflow integration:** The clinician receives an interpretable breakdown showing which factors contributed to the diagnosis and uncertainty level. The system supports counterfactual analysis - clinicians can ask "how would the diagnosis change if only the language component was considered?" This transparency allows clinicians to apply their judgment and understand whether the model's reasoning aligns with their clinical assessment.

**Validation approach:** The most critical element for real-world deployment is accurate population base rates, which can be compiled from epidemiological studies or risk-adjusted clinical data. Our Spanish-language validation demonstrated that providing the correct local prevalence rate (45%) allowed successful adaptation to a completely new population, improving accuracy from 46% to 62% without requiring new training data.

## 4 Data

### 4.1 Measuring Subtle Linguistic Symptoms

Traditional speech analysis focuses on easily quantifiable features like word count, but many clinically recognized symptoms - such as spatial confusion or narrative coherence loss - are nearly impossible to measure computationally.

**Our solution:** We developed a privacy-preserving method using AI to create training data for 66 different linguistic symptoms. By generating 100 synthetic "Cookie Theft" scenarios and having AI rewrite them to exhibit specific AD/ADRD patterns, we created targeted classifiers for symptoms like pronoun overuse and conceptual fixation.

**Research implication:** This approach cost \$25 and the dataset is now freely available on HuggingFace. This experiment provides a template for measuring many linguistic symptoms that experts can describe but tools cannot quantify.

### 4.2 Establishing Healthy Baselines

We used Mozilla's Common Voice dataset - 250+ hours from healthy older adults - to establish robust normative baselines for voice quality analysis.

**Research implication:** Freely available voice datasets can supplement small clinical control groups. Future studies should leverage public datasets rather than relying solely on limited clinical samples.

### 4.3 All Data is Helpful!

Our hierarchical approach demonstrates how even small, biased datasets can contribute valuable information without corrupting the overall model. In Phase 2, the Yu dataset (only Mandarin-speaking MCI patients)

created unrealistic confidence, but our approach incorporates its valuable MCI information while preventing population bias through dataset-specific scaling factors.

## 5 Contributing Factors

### 5.1 Clinical Decision Hierarchy

| Component        | Contribution | Clinical Role            |
|------------------|--------------|--------------------------|
| Voice patterns   | Strongest    | Primary screening signal |
| Demographics     | Moderate     | Baseline risk adjustment |
| Language content | Weakest      | Specialist consultation  |

**Voice patterns dominate:** In 60% of cases, acoustic features drive decisions, aligning with clinical observations that motor changes often precede language deterioration.

**Language as specialist:** Linguistic features drive only 8% of cases - the most complex patients where acoustic signals are ambiguous. When this occurs, confidence drops 32%, correctly flagging cases for expert review.

### 5.2 Clinical Safety Through Uncertainty

The model stratifies patients into actionable categories:

- **High confidence (27%):** Automated screening appropriate
- **Moderate confidence (43%):** Standard decision support
- **Low confidence (30%):** Specialist evaluation required

### 5.3 Research Priorities

**All data contributes value:** Our approach demonstrates that even healthy speech data (like Mozilla Common Voice) improves AD/ADRD detection by establishing robust normative baselines. Prioritize collecting diverse speech samples from underrepresented populations, particularly older Latino and African American speakers.

**Expert-validated synthetic data:** Collaborate with neuropsychologists to validate AI-generated training examples in our synthetic data, enabling detection of previously unmeasurable symptoms like "tangential speech."

**Longitudinal validation:** The long-term goal should be collecting repeat samples from the same individuals over time to track how speech markers change as disease progresses, potentially establishing speech analysis as an early biomarker comparable to neuroimaging.

## 6 References

Here are the citations formatted in APA style.

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